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     ****************************************************
     *                                                  *
     *           STAAD.Pro V8i SELECTseries2            *
     *           Version  20.07.07.32                   *
     *           Proprietary Program of                 *
     *           Bentley Systems, Inc.                  *
     *           Date=    DEC  7, 2011                  *
     *           Time=    16:21:26                      *
     *                                                  *
     *      USER ID: RNS _ASSOCIATES                    *
     ****************************************************
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     1. STAAD SPACE DXF IMPORT OF 20111202 JRC PIG AREA STAIRCASE.DXF
INPUT FILE: 20111207 STAIRCASE PIG STATION.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 03-DEC-11
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 0 0 1; 3 0 1.3065 0; 4 0 1.3065 1; 5 0 2.613 0; 6 0 2.613 1
     9. 7 0 4.0065 0; 8 0 4.0065 1; 9 0 5.4 0; 10 0 5.4 1; 11 0.5 5.4 0; 12 0.5 5.4 1
    10. 13 1 5.4 0; 14 1 5.4 1; 15 1.5 0 0; 16 1.5 0 1; 17 1.5 1.3065 0
    11. 18 1.5 1.3065 1; 19 1.5 2.613 0; 20 1.5 2.613 1; 21 1.5 4.0065 0
    12. 22 1.5 4.0065 1; 23 1.5 5.4 0; 24 1.5 5.4 1; 25 5.1 4E-016 0; 26 5.1 4E-016 1
    13. 27 5.1 1.3065 0; 28 5.1 1.3065 1; 29 5.1 2.613 0; 30 5.1 2.613 1
    14. 31 5.6 2.613 0; 32 5.6 2.613 1; 33 6.1 4E-016 0; 34 6.1 4E-016 1
    15. 35 6.1 1.3065 0; 36 6.1 1.3065 1; 37 6.1 2.613 0; 38 6.1 2.613 1
    16. 39 9.475 4E-016 0; 40 9.475 4E-016 1
    17. MEMBER INCIDENCES
    18. 1 1 3; 2 3 2; 3 4 1; 4 2 4; 5 4 3; 6 3 5; 7 5 4; 8 6 3; 9 4 6; 10 5 6; 11 5 7
    19. 12 7 6; 13 8 5; 14 6 8; 15 8 7; 16 7 9; 17 9 8; 18 10 7; 19 8 10; 20 9 10
    20. 21 9 11; 22 10 12; 23 11 12; 24 3 17; 25 4 18; 26 5 19; 27 6 20; 28 7 21
    21. 29 8 22; 30 11 13; 31 12 14; 32 13 14; 33 13 23; 34 14 24; 35 17 15; 36 17 16
    22. 37 18 15; 38 18 16; 39 18 17; 40 19 17; 41 19 18; 42 20 17; 43 20 18; 44 19 20
    23. 45 21 19; 46 21 20; 47 22 19; 48 22 20; 49 22 21; 50 23 21; 51 23 22; 52 24 21
    24. 53 24 22; 54 23 24; 55 23 29; 56 24 30; 57 27 25; 58 27 26; 59 28 25; 60 28 26
    25. 61 28 27; 62 29 27; 63 29 28; 64 30 27; 65 30 28; 66 29 30; 67 29 31; 68 30 32
    26. 69 27 35; 70 28 36; 71 31 32; 72 31 37; 73 32 38; 74 35 33; 75 35 34; 76 36 33
    27. 77 36 34; 78 36 35; 79 37 35; 80 37 36; 81 38 35; 82 38 36; 83 37 38; 84 37 39
    28. 85 38 40
    29. DEFINE MATERIAL START
    30. ISOTROPIC STEEL
    31. E 2.05E+008
    32. POISSON 0.3
    33. DENSITY 76.8195
    34. ALPHA 1.2E-005
    35. DAMP 0.03
    36. END DEFINE MATERIAL
    37. MEMBER PROPERTY JAPANESE
    38. 1 4 6 9 11 14 16 19 35 38 40 43 45 48 50 53 57 60 62 65 74 77 79 -
    39. 82 TABLE ST H150X150X7
    40. 5 10 15 20 TO 22 24 TO 31 33 34 39 44 49 54 61 66 TO 70 72 73 78 -
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    41. 83 TABLE ST H150X150X7
    42. 55 56 84 85 TABLE ST C200X80X7.5
    43. 23 32 71 TABLE ST C100X50X5
    44. 2 3 7 8 12 13 17 18 36 37 41 42 46 47 51 52 58 59 63 64 75 76 80 -
    45. 81 TABLE ST C100X50X5
    46. CONSTANTS
    47. BETA 180 MEMB 55 84
    48. MATERIAL STEEL ALL
    49. SUPPORTS
    50. 1 2 15 16 25 26 33 34 PINNED
    51. 39 40 FIXED BUT MY MZ KMX 0.00099
    52. MEMBER RELEASE
    53. 5 10 15 20 39 44 49 54 61 66 78 83 START MY MZ
    54. 5 10 15 20 39 44 49 54 61 66 78 83 END MY MZ
    55. 55 56 START MY MZ
    56. 55 56 END MY MZ
    57. 84 85 START MY MZ
    58. 23 32 71 START MY MZ
    59. 23 32 71 END MY MZ
    60. MEMBER TENSION
    61. 2 3 7 8 12 13 17 18 36 37 41 42 46 47 51 52 58 59 63 64 75 76 80 81
    62. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD: DL
    63. SELFWEIGHT Y -1
    64. MEMBER LOAD
    65. *CHEQUERED PLATE LOAD 0.5KN/M^2 X 0.5M = 0.25 KN/M
    66. 55 56 84 85 UNI GY -0.25
    67. *CHEQUERED PLATE LOAD 0.5KN/M^2 X 0.5M = 0.25 KN/M
    68. 23 32 71 UNI GY -0.25
    69. *CHEQUERED PLATE LOAD 0.5KN/M^2 X 0.25M = 0.125 KN/M
    70. 20 54 66 83 UNI GY -0.125
    71. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD: LL
    72. MEMBER LOAD
    73. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M
    74. 55 56 84 85 UNI GY -2.5
    75. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M
    76. 23 32 71 UNI GY -2.5
    77. *LIVE LOAD 5KN/M^2 X 0.25M = 1.25 KN/M
    78. 20 54 66 83 UNI GY -1.25
    79. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD X: WLX
    80. *WIND LOAD 0.85KN/M^2 X 0.15M = 2.5 KN/M X 0.0120 = 0.017KN/M
    81. MEMBER LOAD
    82. 1 4 TO 6 9 TO 11 14 TO 16 19 20 35 38 TO 40 43 TO 45 48 TO 50 53 TO 57 60 -
    83. 61 TO 62 65 66 74 77 TO 79 82 TO 85 UNI GX -0.17
    84. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD Z: WLZ
    85. *WIND LOAD 0.85KN/M^2 X 0.15M = 2.5 KN/M X 0.0120 = 0.017KN/M
    86. MEMBER LOAD
    87. 1 4 6 9 11 14 16 19 21 22 24 TO 31 33 TO 35 38 43 45 48 50 53 55 TO 57 60 -
    88. 62 65 67 TO 70 74 77 79 82 84 85 UNI GZ -0.17
    89. LOAD 5 LOADTYPE NONE  TITLE NOTIONAL LOAD X: NLX
    90. MEMBER LOAD
    91. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
    92. 55 56 84 85 UNI GX -0.0375
    93. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
    94. 23 32 71 UNI GX -0.0375
    95. *LIVE LOAD 5KN/M^2 X 0.25M = 1.25 KN/M X 0.015 = 0.0188KN/M
    96. 20 54 66 83 UNI GX -0.0188
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 97. LOAD 6 LOADTYPE NONE  TITLE NOTIONAL LOAD Z: NLZ
 98. MEMBER LOAD
 99. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
100. 55 56 84 85 UNI GZ -0.0375
101. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
102. 23 32 71 UNI GZ -0.0375
103. *LIVE LOAD 5KN/M^2 X 0.25M = 1.25 KN/M X 0.015 = 0.0188KN/M
104. 20 54 66 83 UNI GZ -0.0188
105. LOAD COMB 7 1.0DL + 1.0LL
106. 1 1.0 2 1.0
107. LOAD COMB 8 1.0DL + 1.0LL +1.0WLX
108. 1 1.0 2 1.0 3 1.0
109. LOAD COMB 9 1.0DL +1.0LL + 1.0WLZ
110. 1 1.0 2 1.0 4 1.0
111. LOAD COMB 10 1.0DL + 1.0LL +1.0NLX
112. 1 1.0 2 1.0 5 1.0
113. LOAD COMB 11 1.0DL +1.0LL + 1.0NLZ
114. 1 1.0 2 1.0 6 1.0
115. LOAD COMB 12 1.4DL +1.6LL
116. 1 1.4 2 1.6
117. LOAD COMB 13 1.2DL + 1.2LL +1.2WLX
118. 1 1.2 2 1.2 3 1.2
119. LOAD COMB 14 1.2DL +1.2LL + 1.2WLZ
120. 1 1.2 2 1.2 4 1.2
121. LOAD COMB 15 1.2DL + 1.2LL +1.2NLX
122. 1 1.2 2 1.2 5 1.2
123. LOAD COMB 16 1.2DL +1.2LL + 1.2NLZ
124. 1 1.2 2 1.2 6 1.2
125. PERFORM ANALYSIS
```

P R O B L E M   S T A T I S T I C S

```
NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =    40/    85/    10

      SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH=    14/     8/     48 DOF
TOTAL PRIMARY LOAD CASES =     6, TOTAL DEGREES OF FREEDOM =    210
SIZE OF STIFFNESS MATRIX =        11 DOUBLE  KILO-WORDS
REQRD/AVAIL. DISK SPACE  =      12.3/   >2000  MB
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```
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=    1


**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=    2
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**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=    3


**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=    4


**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=    5


**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=    6


126. LOAD LIST 7 TO 11
127. PRINT JOINT DISPLACEMENTS ALL
```